KARTHIK TEJAS BASAVARAJAPPA

Boston, MA | basavarajappa.k@northeastern.edu | github.com/Tejas12972 | linkedin.com/in/tejasb12

EDUCATION

Northeastern University | Boston, MA **Expected May 2029**

Candidate for Bachelor of Science in Computer Engineering and Computer Science

Relevant Coursework: Program Design & Implementation 1, Discrete Structures, Fundamentals of Networks (Circuit Analysis), Cornerstone of Engineering 1 & 2, Differential Equations & Linear Algebra, Physics 1 & 2, Calculus 1 & 2

TECHNICAL SKILLS

Hardware & Embedded: Arduino, ESP32, Raspberry Pi, Breadboarding & Soldering, Oscilloscope, Multimeter, Function Generator, Bench Power Supply, UART/I²C/SPI, GPIO/PWM

Languages: C, Python, PyTeal, Java, MATLAB, SQL, TypeScript/JavaScript, HTML/CSS

CAD & Simulation: SolidWorks, AutoCAD, MATLAB/Simulink

Frameworks & Libraries: React.js, Next.js, Node.js, Flask, FastAPI, Streamlit, TensorFlow, Pandas, NumPy, SciPy

Tools & Platforms: Git, GitHub Actions, Docker, AWS, MongoDB, SQLite, pytest, Linux/Unix CLI

WORK EXPERIENCE

XalteraMedia | Andover, MA **Jan 2024 – Present**

Co-Founder

- Build client-acquisition systems end to end for service businesses: Meta ads, Next.js sites, GoHighLevel CRM, booking
- Automate the lead pipeline in GoHighLevel: sub-5-minute response, qualification, recovery, attribution
- Develop voice and text agents that answer inbound calls and qualify leads 24/7 without staff
- Integrate speech-to-text, an LLM response layer, and ElevenLabs voice into one call pipeline

CodeWiz | Reading, MA **Feb 2025 – Jul 2025**

Programming & Robotics Coach

- Coached a FIRST LEGO League team to the MA state title and a 2025 World Festival bid
- Directed the full competition cycle: robot design, autonomous programming, and judging prep
- Taught students to isolate sensor, motor, and code faults on LEGO SPIKE Prime robots in Python
- Mentored students in Python, Java, and Lua, from first program through competition level

Abris.io | Andover, MA **Jun 2022 – Aug 2023**

Software Engineering Intern

- Designed and tested conversation flows for Pizza Pal, an AI host in Uprate.AI's demo suite
- Identified and fixed faulty responses through systematic testing of flows and menu knowledge
- Programmed PyTeal (Python) smart contracts for Licit, an Algorand asset-verification platform
- Developed Algorand on-chain logic storing document hashes, enabling tamper-proof verification in seconds

TECHNICAL PROJECTS

ELISA Analysis Pipeline **Sep 2026**

- Built a Python/Pandas pipeline turning raw 96-well ELISA plate reads into QC-checked concentrations and stats
- Fitted weighted 4PL/5PL curves in SciPy that recover public IL-6 standards at 94–102%, vs. 28–142% for linear fits
- Shipped a Streamlit app that runs in-browser via WebAssembly; 135 pytest tests gated by GitHub Actions

Options Analytics Platform **Sep 2026**

- Developed a Python/FastAPI options analytics platform computing Black-Scholes greeks and implied volatility
- Engineered a daily Pandas snapshot pipeline persisting implied-volatility history to SQLite for IV rank
- Achieved 97% analytics coverage with a 547-test offline pytest suite gated by GitHub Actions

Phone Box **Aug 2026 – Sep 2026**

- Built an ESP32-S3 phone lockbox in Circuit Python with a React Native BLE app and a Firebase web dashboard
- Debugged servo brownouts and PWM drift on the bench; moved the servo to the battery rail
- Wired a touchscreen, servo latch, and MAX17043 I²C fuel gauge; designed two SolidWorks enclosure revisions

RecycleRight **Mar 2025**

- Engineered a Flask app that classifies waste from photo uploads with a TensorFlow model
- Integrated GPT-4o analysis for unclear items, MongoDB storage, and Google Maps recycling-center lookup

LEADERSHIP & VOLUNTEER EXPERIENCE

Eagle Scout | Boy Scouts of America, Andover, MA, Troop 77 **2016 – 2025**

- Led an Eagle Scout project building raised-bed gardens for a pantry that feeds 300+ families